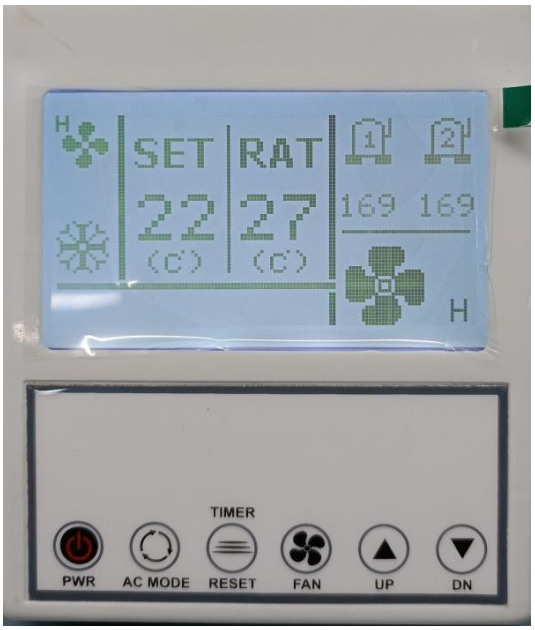
SUMAN ELECTRONICS

Main Screen / Default Screen:



In Main Screen, following parameters are displayed:

1. Return Air Temperature (RAT)
2. Set Temperature (SET)
3. Operating Mode (Cool/Fan/Off)
4. Fan Speed Status (High, Medium, Low)
5. Compressors and Fan On / Off Status
6. Compressor Recycle Time in seconds
7. Faults

- In the above screen, Fan Speed is high. It is shown at right side bottom of screen.
- Operating Mode is Cool. It is shown at left hand side of screen.
- Compressors are in recycle time. So, Recycle Time is flashed below respective compressor symbol. Compressor symbols and recycle time is shown at right hand side top of screen. If Compressor is in LP Bypass time, this LP Bypass time is shown in steady manner below respective compressor.
- Set Temperature (SET) and Return Air Temperature (RAT) are displayed in middle of screen.
- Faults are displayed at the bottom side of screen.

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Below is the screen where both compressors are ON and Fan Speed is Medium:

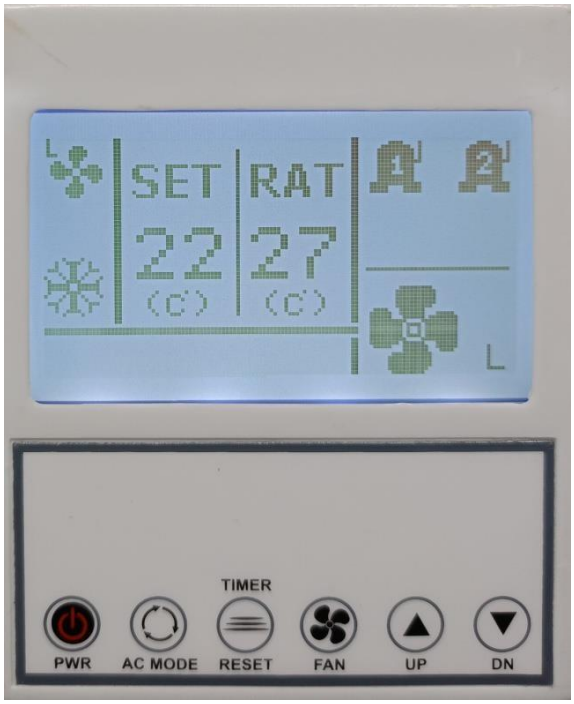


In the following screen, Fan High speed is shown with both compressors ON:

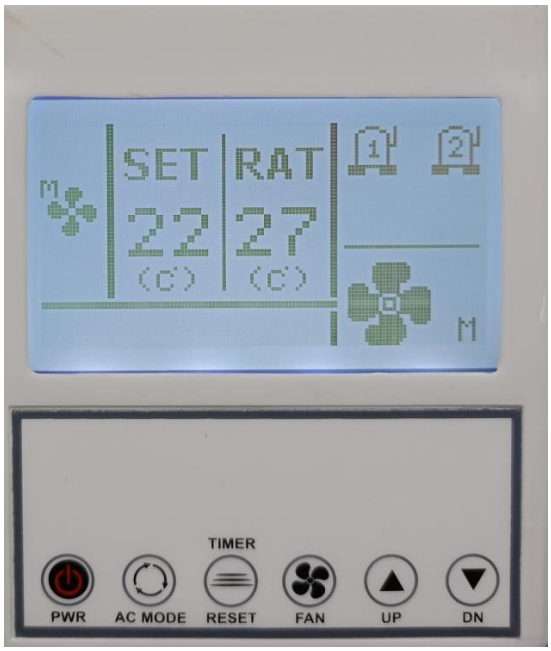


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In the following screen, Fan Low speed is shown when both compressors are ON:

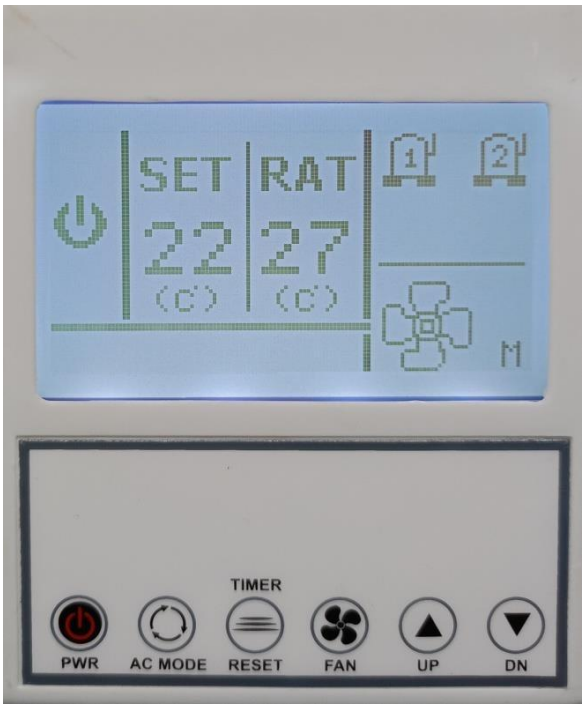


In the following screen, Operating Mode is Fan. Fan Medium speed is shown below. As the operating Mode is Fan, both compressors are off as shown below:

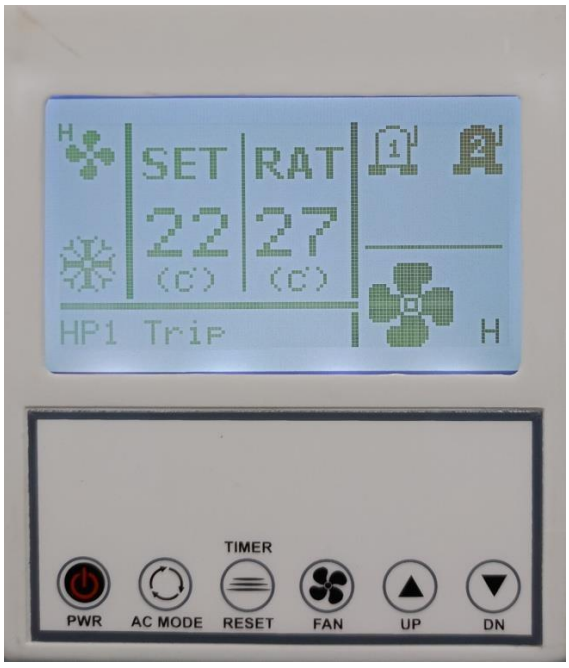


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Following screen shows Controller in OFF Mode. We can see both compressors are off. Fan is also off. Off Mode symbol is seen:



Following screen displays HP1 Fault. Fault is shown flashing at the bottom of screen as follows. Similarly, other faults and reset condition are also shown at the bottom of screen:



SUMAN ELECTRONICS

Touch Key Functions:

1. Power Key:
   Power Key is use to toggle controller mode between ON and OFF.

2. AC Mode Key:
   If AC Mode Key is pressed continuously for 3 seconds, operating mode is changed from Cool to Fan or Fan to Cool mode.

3. Timer/Reset Key:
   This key has 2 functions, namely resetting faults & for current setting.

4. Fan Speed Key:
   This key is used to change Fan speed from high to Medium to Low and again to High.

5. Up/Dn Key:
   These 2 keys are used for changing Set temperature value, address setting, Current setting, Baud Rate Setting, displaying parameters like supply air temperature, Humidity values (2Nos), compressor current.

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Address Setting:

Address setting of the controller can be done within 15 seconds after switching on the power to the controller. For going to Address Setting Mode, following is the procedure:

1. Press AC Mode Key & Reset key simultaneously & continuously for about 5 seconds. Controller will enter into Address Setting menu.
2. It will display current Slave Address blinking.
3. Press Up/Dn key for changing the address. Slave Address can be set from 1 to 247.
4. Press Reset Key to set the Slave Address to the changed value.

Following is the screen for Address Setting:



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Baud Rate Setting:

Baud Rate Setting can be done only after 15 seconds after power on. This menu is password protected. Password is 8010. Allowed Baud Rates are 9600 bps and 19200 bps.

Following is the procedure for Baud Rate setting:

1. Press AC Mode key & Reset key simultaneously & continuously for 5 seconds. Controller will enter into password menu. Release AC Mode Key & Reset key.
2. Password is seen as "----". Password digits can be changed using Up/Dn keys. Up Key is used for changing digit value from 0 to 9. Dn key is used to shift from one digit to other.
3. Press Reset Key once. If password entered is incorrect, Display will show "Invalid Password! Access Denied!" screen for 3 seconds and will return to Main Screen.
4. If Password is correct, display will show Current Baud Rate Value blinking.
5. Using Up/Down keys, baud rate can be changed. For setting Baud rate to desired value, Press reset key. Baud Rate can be 9600 bps or 19200 bps.
6. New Baud Rate is set & Display will show Message "New Baud Rate Set! Please Restart System" on screen. Controller will Halt and wait for user to restart the system.

   Following is the screen for Baud Rate Setting:



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Temperature Setting:

Set temperature can be changed from 19 to 32 Deg. Celsius. Following is the procedure for temperature setting:

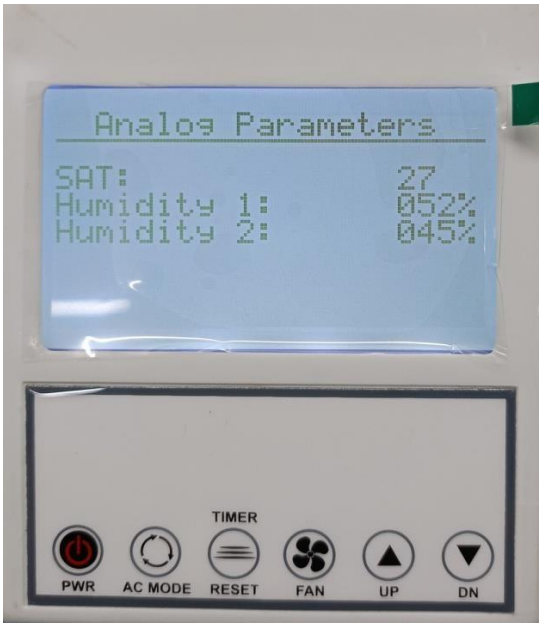
1. Press Up or Dn key to enter into temperature setting menu. Controller will start blinking set temperature value.
2. Press Up/Dn key to change set temperature value.
3. The changed value will blink for 5 seconds. After 5 seconds, this new set temperature value is saved.

Parameter Display Menu:

In main screen, Return Air Temperature (RAT) and Set Temperature (SET) are displayed. There are other parameters in the system. Following is the procedure for viewing parameters in the system:

1. Press Up Key continuously for 3 seconds. Controller will enter in parameter display menu.
2. All other parameters are displayed on screen. The parameters are namely
   a. Supply Air Temperature
   b. Humidity 1
   c. Humidity 2
   d. Compressor 1 Current(If compressor sensing is enabled & ODU is present)
   e. Compressor 2 Current (If compressor sensing is enabled & ODU is present & no. of compressors > 1).

Following is the Parameter Display Screen:



In the above screen, Current is not shown as ODU is not present. This Parameter Display Menu will be displayed for 10 seconds. After that, controller will return back to Main Screen.

Messages on screen:

Following are the various messages on the screen & their description:

Sr. No	Message on screen	Description
1	BF SPPR	3 Phase Fan SPPR fault if the Fan is 3 phase in IDU ODU system.
2	SPPR Trip	System SPPR Fault
3	HPLP1 / HPLP2	HPLP fault of Compressor 1 / Compressor 2
4	HP1 Trip / HP2 Trip	HP fault of Compressor 1 / Compressor 2
5	LP1 Trip / LP2 Trip	LP fault of Compressor 1 / Compressor 2
6	RST	Fault resetting condition is present.
7	SAT TH Op	Supply Air Temperature Thermistor Open
8	RAT TH Op	Return Air Temperature thermistor open
9	H1 Open	Humidity 1 Sensor Open
10	H2 Open	Humidity 2 Sensor Open
11	C1 / C2 UL 150(Recycle Time)	Compressor 1 / Compressor 2 Underload fault with compressor recycle time in case fault is going to get cleared after recycle time.
12	C1 / C2 UL	Compressor 1 / Compressor 2 Underload fault
13	C1 / C2 OL 150(Recycle Time)	Compressor 1 / Compressor 2 Overload fault with compressor recycle time in case fault is going to get cleared after recycle time
14	C1 / C2 OL	Compressor 1 / Compressor 2 Overload fault
15	C1 / C2 SPPR	Compressor 1 / Compressor 2 SPPR fault
16	C1 / C2 J2ER	Compressor 1 / Compressor 2 J2ER fault. This fault is declared when compressor is off but its current value is present.
17	LPb1 / LPb2 100 (LP Bypass time in seconds)	LP Bypass time of Compressor 1 / Compressor 2. LPB1/2 string is blinking while LP Bypass time is steady display.
18	Invalid Password! Access Denied!	Wrong Password Entered
19	Comm. Err	Communication Error between Handset & IDU / IDU & ODU
20	PC Mode!	IDU is in PC (Remote) Mode. Thus, Operating Mode, Set Temperature can be set from PC and not from LCD Handset